

Cleaning and Preparing Data

150 minutes · 6 self-check questions

Jesse Anak · Institute of Computational Biology and Bioinformatics

Finding what is wrong with a real dataset — impossible values, duplicates, inconsistent categories — and building the derived variables the analysis actually needs, all as a script you can rerun.

Learning objectives

- Screen every variable for implausible and out-of-range values
 - Detect and resolve duplicate records
 - Recode, collapse and reverse-score variables correctly
 - Compute derived variables such as BMI and scale totals
 - Decide whether an outlier is an error or a real observation
 - Keep cleaning reproducible by scripting it rather than editing cells
-

First look at a new dataset

- Check dimensions against expectation: does the row count match the number of questionnaires?
 - Run a frequency table for every categorical variable and a min/max for every continuous one
 - In R: `summary(dat)` then `table(dat$sex, useNA = "ifany")` for each factor
 - In SPSS: Analyze > Descriptive Statistics > Frequencies, ticking Minimum and Maximum
 - Look for categories that should not exist — a sex coded 3, an education level of 7
 - Look for values at the boundaries: an age of 0, a blood pressure of 999
-

Impossible, implausible and duplicate

- Impossible: outside the physically possible range — a diastolic BP of 400, a date of birth in the future
- Implausible: possible but unlikely enough to warrant checking — a 3-year-old with three children
- Cross-variable checks catch what single-variable checks miss: pregnant men, ages inconsistent with dates of birth
- Duplicates may be exact (entered twice) or partial (same participant, two IDs)
- Check for duplicates on the identifier and on a combination of key fields
- Resolve against the paper form where possible; where not, document the decision

IN PRACTICE

Return to the source before you "correct" anything. A value that looks impossible is sometimes a real and interesting observation, and sometimes a data entry slip — and only the original form tells you which.

Recoding and deriving

- Collapse categories only with a reason, and state it: small cell counts, or a clinically meaningful threshold
- Reverse-score negatively worded items before summing a scale, or the total is meaningless
- For a 1-5 item, the reversed value is 6 minus the original
- Compute scale totals only when the required number of items is present, and say how you handled partial scales
- Derived variables get their own names and codebook entries — never overwrite the source variable
- BMI in R: `dat$bmi <- dat$weight_kg / (dat$height_m)^2`
- Banding continuous variables loses information; do it for presentation, not for the primary analysis

Outliers

- An outlier is an observation far from the others; it is not automatically an error
- Detect with boxplots, standardised scores beyond about ± 3 , or simply sorting the variable
- Ask first whether it is plausible: a systolic BP of 210 is high but real; 2100 is a typing slip
- If it is an error and the true value is recoverable, correct it and document it
- If it is a genuine extreme value, keep it — and report a sensitivity analysis with and without it
- Deleting inconvenient observations because they weaken a result is data manipulation, not cleaning

Doing it reproducibly

- Every change goes in a script: R, SPSS syntax, or Stata do-file
- Point-and-click cleaning cannot be checked, cannot be rerun, and cannot be corrected when you find a mistake at step three
- Structure the script: read raw, clean, derive, save analysis dataset
- Never overwrite the raw file; write a separate cleaned file
- Keep a running count so you can report how many records were excluded and why
- Rerunning the whole script from raw data should reproduce the analysis dataset exactly

Worked examples

Six checks that catch most entry errors

A district survey of 600 households arrives as a clean-looking CSV. Before analysing anything, it needs screening.

1. Dimensions: `nrow(dat)` should be 600. It is 603 — three extra rows.
2. Duplicates: `sum(duplicated(dat$id))` returns 3. The same households were entered twice.
3. Ranges: `summary(dat$age_years)` shows a maximum of 199. A typing slip for 19 or 99.
4. Categories: `table(dat$edu)` shows a level 7 that the codebook does not define.
5. Cross-variable: `sum(dat$sex == 1 & dat$pregnant == 1)` returns 2 pregnant men.
6. Missing codes: `min(dat$income)` is -99, so the missing code was imported as a real value and every mean is wrong.

THE POINT

Every one of these passes a casual look at the spreadsheet. Six commands, run before any analysis, catch them all — and the last one would otherwise have silently corrupted every income statistic in the paper.

The outlier that was real

In a study of hypertension, one participant has a systolic pressure of 218 mmHg. It sits far above the rest and weakens the association the student expected. They ask whether to remove it.

1. First: is it possible? 218 mmHg is high but entirely attainable, and this is a hypertension study.
2. Second: check the source. The paper form says 218, recorded twice, with a note that the patient was referred.
3. So it is not an error. It is precisely the kind of participant the study exists to capture.
4. Removing it because it weakens the finding would be data manipulation, not cleaning.
5. Keep it, and run the analysis with and without it as a sensitivity check.
6. Report both: "excluding the participant with SBP 218, the association was 4.1 mmHg (95% CI 0.8 to 7.4) rather than 5.2".

THE POINT

The question is never "does this point help my result" but "is this value real". A sensitivity analysis lets you keep an honest observation and still show it is not driving the conclusion.

Key terms

Range check

Verifying every value of a variable falls within its permitted minimum and maximum.

Cross-variable check

Testing that combinations of values are logically consistent with each other.

Reverse scoring

Inverting negatively worded items so that all items on a scale point the same way.

Derived variable

A new variable computed from existing ones, such as BMI or a scale total.

Outlier

An observation distant from the rest of the distribution; an error or a genuine extreme.

Sensitivity analysis

Repeating an analysis under a different assumption to see whether the conclusion holds.

Activity

You are given a dataset of 600 records. Write down the checks you would run before analysing anything — list them variable by variable, including at least three cross-variable checks. Then write the R or SPSS syntax for two of them.

Further reading

- **Van den Broeck et al. (2005)** — "Data Cleaning: Detecting, Diagnosing, and Editing Data Abnormalities" — Open access, and the clearest framework: screening, diagnosis, treatment.
- **Osborne** — **Best Practices in Data Cleaning** — Practical, with worked examples of what goes wrong.
- **R for Data Science, chapters on data transformation** — Free online; dplyr makes cleaning scripts readable.

Self-check questions

1. A participant is recorded as male and 6 months pregnant. This is caught by:
 - a. A range check on sex
 - b. A cross-variable consistency check
 - c. A test for outliers
 - d. Cronbach's alpha
2. On a 1-5 agreement scale, the reversed value of a response of 2 is:

a. 3

- b. 4
 - c. 5
 - d. 2
3. A systolic blood pressure of 210 mmHg appears in your data. You should:
- a. Delete it as an outlier
 - b. Check it against the source, and if genuine keep it and consider a sensitivity analysis
 - c. Replace it with the mean
 - d. Round it down to the next plausible value
4. The strongest argument for scripting all cleaning rather than editing cells is that:
- a. Scripts run faster
 - b. The whole process can be rerun, checked and corrected, and the raw data stays intact
 - c. SPSS does not allow manual editing
 - d. It uses less memory
5. A total score is computed for participants who answered only 4 of 9 items. The safest approach is to:
- a. Sum whatever is present and treat it as a full score
 - b. Set a minimum number of completed items, apply it consistently, and report the rule
 - c. Always exclude anyone with any missing item
 - d. Replace missing items with zero
6. Which of these is data manipulation rather than cleaning?
- a. Correcting a transcription error against the paper form
 - b. Removing genuine observations because they weaken the association you expected
 - c. Flagging an impossible date of birth for checking
 - d. Documenting that three records were excluded as duplicates
-

Answers

1. **b.** Both values are individually permissible — the impossibility only appears when they are considered together. Range checks alone would pass this record.
2. **b.** For a 1-k scale the reversed value is $(k + 1)$ minus the original, so $6 - 2 = 4$. Forgetting to reverse negatively worded items is one of the most common scale errors and quietly destroys the total.
3. **b.** It is high but entirely possible — this is exactly the kind of participant a study of hypertension exists to capture. Verify it, keep it, and show that the conclusion does not depend on it.
4. **b.** Manual edits are invisible and unrepeatable. If you discover an error in step three, a script lets you fix it and rerun; hand-edited cells mean starting over with no record of what was changed.

5. **b.** Summing four items gives a total that is not comparable to a nine-item total. Common practice is to require a stated proportion complete, prorate or impute the remainder, and say exactly what was done.
6. **b.** The distinction is whether the decision is driven by the data being wrong or by the result being unwelcome. Excluding valid observations to improve a finding is misconduct, however it is described in the write-up.

